

Institute of Actuaries of India

Subject CT3 – Probability & Mathematical Statistics

November 2012 Examinations Indicative Solutions

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other approaches leading to a valid answer and examiners have given credit for any alternative approach or interpretation which they consider to be reasonable

Q1 Let Y = Claim Amount

P (Disability Claim) = 0.35, P (Dismemberment Claim) = 0.65

E (Y | Disability Claim) = 50,000, Var (Y | Disability Claim) = 10,000²

E (Y | Dismemberment Claim) = 30,000, Var (Y | Dismemberment Claim) = 7,000²

[0.5]

The mean of claim amounts per policy = E(Y)

= E (Y | Disability Claim) × P (Disability Claim) + E (Y | Dismemberment Claim) × P (Dismemberment Claim)

= 50,000 × 0.35 + 30,000 × 0.65

= 17,500 + 19,500 = 37,000

[1.5]

E (Y²)

= E (Y² | Disability Claim) × P (Disability Claim) + E (Y² | Dismemberment Claim) × P (Dismemberment Claim)

= (10,000² + 50,000²) × 0.35 + (7,000² + 30,000²) × 0.65

= 910,000,000 + 616,850,000

= 1,526,850,000

[2]

The variance of claim amounts per policy = Var (Y)

= E (Y²) – [E(Y)]²

= 1,526,850,000 – 37,000²

= 157,850,000

[1]

Thus, the standard deviation of claim amounts per policy = 12,564

[1]

[Total 6]

Q2 The mean of the gamma distribution is $\frac{\alpha}{\lambda} = 6.25$

[0.5]

The variance of the gamma distribution is $\frac{\alpha}{\lambda^2} = 15.625$

[0.5]

The mode will be the value of x, which gives the greatest value of the PDF. We can find the mode by differentiating the PDF or (equivalently) the log of the PDF:

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$$

$$\log f(x) = (\alpha - 1) \log x + \alpha \log \lambda - \lambda x - \log \Gamma(\alpha)$$

$$\frac{d}{dx} \log f(x) = \frac{\alpha - 1}{x} - \lambda$$

Setting this equal to zero to obtain the maximum turning point gives:

$$x = \frac{\alpha - 1}{\lambda} = 3.75 \text{ for the mode}$$

[Note: $\frac{d^2}{dx^2} \log f(x) = -\frac{\alpha-1}{x^2} < 0 \Rightarrow \text{maximum}$]

[2.5]

Therefore:

$$\psi = \frac{6.25 - 3.75}{3.95285} = 0.632$$

[0.5]

[Total 4]

Q3 Revised Sample Mean = $(213,200 - 11,000 - 48,000 + 27,500 + 31,500)/12 = (213,200/12)$
= 17,767

[1]

Revised $\Sigma x^2 = 4,919,860,000 - 121,000,000 - 2,304,000,000 + 756,250,000 + 992,250,000$
= 4,243,360,000

[2]

Therefore: Revised Sample Standard Deviation = $\left\{ \frac{1}{11} \left(4,243,360,000 - \frac{213,200^2}{12} \right) \right\}^{1/2}$

$$= \left\{ \frac{1}{11} (455,506,666.7) \right\}^{1/2}$$

$$= \{41,409,696.97\}^{1/2}$$

$$= 6,435.$$

[2]

Comment: There has been no change in the sample mean. However there has been a reduction in the Sample Standard Deviation from 10,144 to 6,435. [1]

The mean remains unaltered as the total salary of the temporary employees who were being replaced is equal to the total salary of the permanent employees who replaced them. The reason for the reduction in standard deviation is that the salaries of the permanent employees who replaced temporary employees are closer to the sample mean. [1]

[Total 7]

Q4 Let X_1 , X_2 and X_3 denote that the driver is in Group A, Group B and Group C respectively.

Therefore:

$$P(X_1) = 22\%$$

$$P(X_2) = 43\%$$

$$P(X_3) = 35\%$$

Let Z be the event where the driver has made a claim following an accident during 12 months.

Therefore:

$$P(Z|X_1) = 11\%$$

$$P(Z|X_2) = 3\%$$

$$P(Z|X_3) = 2\%$$

To calculate percent of the company's policyholders are expected to make a claim after having an accident during the next 12 months

$$P(Z) = \sum_{i=1}^4 P(Z|X_i) \times P(X_i) = 0.11 * 0.22 + 0.03 * 0.43 + 0.02 * 0.35 = 0.0441 \quad [2]$$

To calculate the probability that a driver who made a claim following an accident is from Group A:

$$P(X_1|Z) = \frac{P(Z|X_1) \times P(X_1)}{\sum_{i=1}^4 P(Z|X_i) \times P(X_i)} = \frac{0.11 * 0.22}{0.0441} = 0.5488 \quad [3]$$

[Total 5]

Q5(i) a. From the definition of the MGF:

$$M_X(t) = E[e^{tX}]$$

Differentiating with respect to t gives:

$$M'_X(t) = E[Xe^{tX}]$$

Putting $t = 0$ gives $M'_X(0) = E[Xe^0] = E[X] \quad [1]$

b. Differentiating again gives:

$$M''_X(t) = E[X^2 e^{tX}]$$

Putting $t = 0$ gives $M''_X(0) = E[X^2 e^0] = E[X^2] \quad [1]$

$$\begin{aligned}\text{Var}(X) &= E[X^2] - E^2[X] \\ &= M''_X(0) - (M'_X(0))^2\end{aligned}\quad [1]$$

(ii) $M_Y(t) = E[e^{tY}] = E[e^{t(X_1+X_2)}]$

Since X_1 and X_2 are independent [1]

$$M_Y(t) = E[e^{tX_1}] \times E[e^{tX_2}]$$

$$\therefore M_Y(t) = M_{X_1}(t) \times M_{X_2}(t) \quad [1]$$

(iii) Since J, K and L are independent, the moment-generating function for their sum, Y, equal to the product of the individual moment-generating functions, i.e. [1]

$$M_Y(t) = M_J(t) \times M_K(t) \times M_L(t)$$

$$\begin{aligned}M_Y(t) &= (1 - 2t)^{-3} \times (1 - 2t)^{-2.5} \times (1 - 2t)^{-4.5} \\ &= (1 - 2t)^{-10}\end{aligned}\quad [1]$$

Differentiating with respect to t gives:

$$M'_Y(t) = (-2) \times (-10) \times (1 - 2t)^{-11}$$

$$M'_Y(0) = (-2) \times (-10) \times 1 = 20$$

Differentiating again gives:

$$M''_Y(t) = (-2)^2 \times (-10) \times (-11) \times (1 - 2t)^{-12}$$

$$M'_Y(0) = 4 \times (-10) \times (-11) \times 1 = 440 \quad [2]$$

$$E(Y) = M'_Y(0) = 20$$

$$\text{Var}(Y) = M''_Y(0) - (M'_Y(0))^2 = 440 - 20^2 = 40 \quad [1]$$

[Alternate approaches like computing mean and variance of J, K & L first and then computing aggregate mean and variance or using the form of MGF to argue that Y follows χ^2_{20} distribution and then computing mean and variance of a chi-square distribution will earn credit as well]

[Total 10]

- Q6 (i) The Cramér-Rao Lower Bound result holds under very general conditions except where the range of the distribution involves the parameter, such as the uniform distribution in this case.

This is due to a discontinuity, so the derivative in the formula doesn't make sense. [1]

- (ii) We have $Y = \max_i X_i$.

Since each X_i lies between 0 and θ , the support for Y will be 0 and θ . [1]

For $0 < y < \theta$,

$$\begin{aligned} P[Y < y] &= P\left[\max_i X_i < y\right] \\ &= P\left[\bigcap_{i=1}^n (X_i < y)\right] \\ &= \prod_{i=1}^n P[X_i < y] \quad [\because X_i \text{ s are independent}] \\ &= \prod_{i=1}^n \left[\int_0^y \frac{dx}{\theta} \right] = \left(\frac{y}{\theta}\right)^n \end{aligned}$$

[1]

Thus the probability density function of Y will be:

$$g_Y(y) = \frac{d}{dy} P[Y < y] = \frac{n}{\theta^n} \cdot y^{n-1} \quad \text{for } 0 < y < \theta$$

[1]

Now, for any non-negative real number k ,

$$\begin{aligned} E[Y^k] &= \int_0^\theta y^k \cdot \frac{n}{\theta^n} \cdot y^{n-1} dy \\ &= \frac{n}{n+k} \int_0^\theta \frac{n+k}{\theta^n} \cdot y^{n+k-1} dy \\ &= \frac{n}{n+k} \cdot \frac{\theta^{n+k} - 0}{\theta^n} \end{aligned}$$

$$= \frac{n \theta^k}{n + k}$$

[2]

(iii) Bias of the estimator $\hat{\theta}(c)$ is given as:

$$\begin{aligned} \text{Bias}[\hat{\theta}(c)] &= E[\hat{\theta}(c) - \theta] \\ &= E[cY - \theta] \\ &= c \cdot E[Y] - \theta \\ &= c \cdot \frac{n \theta}{n + 1} - \theta \quad [\text{using the results in (b) with } k = 1] \\ &= \left(\frac{cn}{n + 1} - 1 \right) \cdot \theta \end{aligned}$$

[2]

Mean Square Error of the estimator $\hat{\theta}(c)$ is given as:

$$\begin{aligned} \text{MSE}[\hat{\theta}(c)] &= E[\{\hat{\theta}(c) - \theta\}^2] \\ &= E[\{cY - \theta\}^2] \\ &= E[c^2 Y^2 - 2c\theta Y + \theta^2] \\ &= c^2 \cdot E(Y^2) - 2c\theta E[Y] + \theta^2 \\ &= c^2 \cdot \frac{n \theta^2}{n + 2} - c \cdot \frac{2n \theta^2}{n + 1} + \theta^2 \quad [\text{using (ii) with } k = 1 \text{ \& } k = 2] \end{aligned}$$

[2]

(iv) For $\hat{\theta}(c)$ to be an unbiased estimator of θ , we need: $\text{Bias}[\hat{\theta}(c_u)] = 0$

$$\Rightarrow \left(\frac{c_u n}{n + 1} - 1 \right) \cdot \theta = 0 \quad \Rightarrow \quad c_u = \frac{n + 1}{n}.$$

[1]

(v) In order to minimize $\text{MSE}[\hat{\theta}(c)]$, we need:

$$0 = \frac{d}{dc} \text{MSE}[\hat{\theta}(c)] = \frac{d}{dc} \left[c^2 \cdot \frac{n \theta^2}{n + 2} - c \cdot \frac{2n \theta^2}{n + 1} + \theta^2 \right] = 2c \cdot \frac{n \theta^2}{n + 2} - \frac{2n \theta^2}{n + 1}$$

$$\text{Thus: } c_m = \frac{2n\theta^2}{n+1} \cdot \frac{n+2}{2n\theta^2} = \frac{n+2}{n+1}. \quad [1]$$

$$\left(\text{Note: } \frac{d^2}{dc^2} \text{MSE}[\hat{\theta}(c)] = \frac{d}{dc} \left[2c \cdot \frac{n\theta^2}{n+2} - \frac{2n\theta^2}{n+1} \right] = \frac{2n\theta^2}{n+2} > 0 \Rightarrow \text{minimum} \right) \quad [1]$$

- (vi) In order to minimize error in estimation, it is preferred to opt for an estimator which has lower mean square error among different competing estimators. So here $\hat{\theta}(c_m)$ will be preferred over $\hat{\theta}(c_u)$. [1]

As n becomes large, $\hat{\theta}(c_u) = \frac{n+1}{n}Y \rightarrow Y$. Similarly $\hat{\theta}(c_m) = \frac{n+2}{n+1}Y \rightarrow Y$
 Thus the two estimators become one and the same as $n \rightarrow \infty$. [1]

[Total 15]

- Q7 Let X be the number in the sample who thinks team England will win.

$$\begin{aligned} X &\sim \text{Binomial}(280, 0.40) \\ E[X] &= 280 \times 0.40 = 112 \\ V[X] &= 280 \times 0.40 \times 0.60 = 67.2 \end{aligned} \quad [1]$$

The normal approximation to the binomial gives, using a continuity correction,

$$\begin{aligned} P(X \geq 106) &= P(X > 105.5) \approx 1 - \Phi\left(\frac{105.5 - 112}{\sqrt{67.20}}\right) \\ &= 1 - \Phi(-0.7929) = \Phi(0.7929) = 0.786 \end{aligned} \quad [2]$$

[Total 3]

- Q8 (i) The likelihood function is given by:

$$\begin{aligned} L(\lambda; \underline{x}) &= \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right)^{33} \left(\frac{\lambda^1}{1!} \right)^{11} \left(\frac{\lambda^2}{2!} \right)^6 \left(\frac{\lambda^3}{3!} \right)^4 \left(\frac{\lambda^4}{4!} \right)^5 \left(\frac{\lambda^6}{6!} \right)^1 \left(\frac{\lambda^8}{8!} \right)^2 \left(\frac{\lambda^9}{9!} \right)^1 \left(\frac{\lambda^{11}}{11!} \right)^1 \left(\frac{\lambda^{12}}{12!} \right)^1 \left(\frac{\lambda^{13}}{13!} \right)^1 \\ &= \text{constant} * \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right)^{33} * \lambda^{122} \end{aligned} \quad [2]$$

The log-likelihood function is given by:

$$l(\lambda; \underline{x}) = \ln L(\lambda; \underline{x}) = \text{constant} + 33 * \ln \left(\frac{e^{-\lambda}}{1 - e^{-\lambda}} \right) + 122 * \ln \lambda$$

To obtain the maximum likelihood estimate (MLE) $\hat{\lambda}$ of the parameter λ , we need to set:

$$\frac{d}{d\lambda} l(\lambda; \underline{x}) = 0$$

$$0 = \frac{d}{d\lambda} l(\lambda; \underline{x}) = -33 * \left(\frac{1}{1 - e^{-\lambda}} \right) + \frac{122}{\lambda}$$

[1]

Thus, the MLE $\hat{\lambda}$ satisfies the following equation: **$33 \hat{\lambda} = 122 (1 - e^{-\hat{\lambda}})$**

[1]

$$\left[\text{Note: } \frac{d^2}{d\lambda^2} l(\lambda; \underline{x}) = -33 * \frac{e^{-\lambda}}{(1 - e^{-\lambda})^2} - \frac{122}{\lambda^2} < 0 \Rightarrow \text{maximum} \right]$$

[1]

(ii) MLE $\hat{\lambda} = 3.6$.

In order to test for the null hypothesis that the number of breakages in a damaged chromosome follows the given probability distribution, we need to perform a chi-square test.

We need to group some of the cells in the frequency distribution so that the expected count of number of breakages is at least 5 in all the cells.

[1]

The resultant frequency table and the χ^2 calculations are as below:

Number of breakages	Observed Number of chromosomes	Expected Number of chromosomes	χ^2
≤ 2	17	9.34	6.27
3	4	7.21	1.43
4	5	6.49	0.34
≥ 5	7	9.96	0.88
Total	33	33.00	8.92

[3]

Here: We have combined the cells with number of breakages 1 & 2 into " ≤ 2 " and with number of breakages more than 4 into " ≥ 5 ".

[1]

The χ^2 critical values for degrees of freedom 2 (= # cells – 1 – 1) taken from the table are:

P	CritVal
0.05%	15.20
0.10%	13.82
0.50%	10.60
1.00%	9.21
5.00%	5.99

This indicates that the null hypothesis can be rejected at 5% level of significance. However it can be accepted at 1% or lower level of significance. In fact the p-value for this test is 0.012. Thus, we can conclude that there **may not be** strong enough evidence to agree that the number of breakages in a damaged chromosome follows the given probability distribution. [2]

[Total 12]

Q9 (i) The assumptions required for one-way analyses of variance (ANOVA) are:

- The populations must be **normal**
- The populations have a **common variance**
- The observations are **independent**. [1]

The sample variance observed for the four rates appear very different from each other. Thus, we can clearly see that the assumption that the underlying populations have a common variance assumption will not hold for the data as they are. [1]

(ii) For the transformation $x \rightarrow \sqrt{x}$, the value of sample mean for rate 1 will be:

$$\frac{1}{3}(\sqrt{29} + \sqrt{13} + \sqrt{21}) = 4.52 \quad [1]$$

For the transformation $x \rightarrow \log_e x$, the value of sample variance for rate 2 will be

$$\frac{1}{2}[(\log_e 180 - 4.827)^2 + (\log_e 90 - 4.827)^2 + (\log_e 120 - 4.827)^2] = 0.1213 \quad [2]$$

(iii) The scientist was correct in asserting that the $\log_e$ transformation must be done before carrying out a one-way ANOVA as for this transformation it can be claimed that the assumption of common variance for the underlying population holds. [1]

To justify this, a quick check can be done on the ratio of maximum to minimum sample variance among the four rates data. A smaller ratio and close to 1 would indicate that the variances are close enough which in turn implies that the assumption of common variance for the underlying population holds.

Variance	x	$\sqrt{x}$	$\log_e(x)$	$1/x$
Min	64	0.79	0.1213	0.0000004
Max	63,300	23.96	0.1861	0.0004721
Ratio	989.06	30.17	1.53	1,268.14

Clearly, the transformation $\log_e x$ produces the **minimum** ratio of maximum to minimum observed sample variance and that too close to 1. [1]

[Note this is one of the approaches to make this argument. The idea is to establish for which transformation we can claim that the variances are “statistically equal” without actually carrying out any hypothesis test. The students can use alternate approaches to argue as long as they are valid. One such argument can be based upon using suitable dot plots]

(iv) We will perform an ANOVA on the $\log_e x$ data. We would assume the following model:

$$Y_{ij} = \mu + \tau_i + e_{ij}, \quad i = 1, 2, 3, 4; \quad j = 1, 2, 3$$

Here:

- Y_{ij} is the $\log_e$ transformed value of the j^{th} observation of the number of germinations per square foot observed when the i^{th} rate was applied
- μ is the overall population mean
- τ_i is the deviation of the i^{th} rate mean such that $\sum \tau_i = 0$
- e_{ij} are the independent error terms which follows Normal distribution with mean 0 and common unknown variance σ^2 [1]

We have already argued that we can assume equal underlying variances for this transformation. So, all requisite assumptions hold here.

For ANOVA, the null hypothesis being tested here is:

$$H_0: \tau_i = 0, \quad i = 1, 2, 3, 4 \quad \text{against} \quad H_1: \tau_i \neq 0 \text{ for at least one } i \quad [1]$$

To carry out the ANOVA, we must first compute the Sum of Squares. We have the following table using the information given in the question:

Rate	$\log_e(x)$		
	Mean	Variance	Y_i^2
1	2.992	0.163	80.582
2	4.827	0.121	209.678
3	5.716	0.186	294.053
4	6.575	0.148	389.060
	20.110	0.618	973.373

Here: $Y_{i.}^2 = (\sum_{j=1}^3 Y_{ij})^2 = (3 * Mean_i)^2$

Now:

- $SS(\text{Rate}) = \sum_{i=1}^4 \frac{Y_{i.}^2}{3} - \frac{Y_{..}^2}{12} = \frac{973.373}{3} - \frac{(3*20.110)^2}{12} = 21.153$
 - $SS(\text{Residuals}) = \sum_{i=1}^4 \left\{ \sum_{j=1}^3 (Y_{ij} - \bar{Y}_{i.})^2 \right\} = \sum_{i=1}^4 \{2 * \text{Variance}_i\} = 2 * 0.618 = 1.236$
- [3]

The ANOVA table is as follows:

Source of Variation	d.f.	Sum of Squares	Mean Squares	F
Rates	3	21.153	7.051	45.638
Residuals	8	1.236	0.155	
Total	11	22.389		

[2]

The 1% critical value for F (3, 8) distribution is 7.951.

Given the observed F statistic value is much larger than this, we can state the p-value for this test is almost near to zero or in other words there is overwhelming evidence against the null hypothesis H_0 . Thus it can be concluded that the underlying means are not equal.

[1]

- (v) A 95% confidence interval for the i^{th} rate mean (in the chosen transformed scale) is given by:

$$\bar{y}_{i.} \pm t_{8;0.975} * \frac{\hat{\sigma}}{\sqrt{3}} \text{ for } i = 1, 2, 3, 4$$

[2]

Here: $t_{8; 0.975}$ is the 97.5% critical point of a t-distribution with 8 degrees of freedom and equals 2.306.

The common error variance σ^2 is estimated as:

$$\hat{\sigma}^2 = \frac{\text{Residual Sum of Squares}}{8} = 0.155 \quad [\text{from the ANOVA table earlier}] \quad [1]$$

The 95% confidence intervals (in transformed and original scale after applying the transformation $CI \rightarrow e^{CI}$) are tabulated in the following table.

Rate	Observed Mean	95% Confidence Interval (Transformed Scale)		95% Confidence Interval (Original Scale)	
		LCI	UCI	LCI	UCI
1	2.992	2.469	3.516	11.810	33.635
2	4.827	4.303	5.350	73.954	210.623
3	5.716	5.193	6.239	179.950	512.505
4	6.575	6.052	7.098	424.770	1,209.763

[2 + 2]

[Total 22]

Q10 (i) The missing entries can be computed as below:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 60 - (-13.958) * 5.71 = 139.700$$

$$s.e(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{S_{xx}}} = \frac{18.2834}{\sqrt{(10-1) * 13.425}} = 1.663$$

$$t(H_0: \beta = 0) = \frac{\hat{\beta} - 0}{\hat{\sigma}/\sqrt{S_{xx}}} = \frac{-13.958}{1.663} = -8.39$$

$$R^2 = 1 - \frac{(n-2) \hat{\sigma}^2}{S_{yy}} = 1 - \frac{8 * (18.2834)^2}{(10-1) * 2912.667} = 89.8\%$$

[1 + 1 + 1 + 1]

Here we assume that the error terms are independent and normal variables with mean 0 and unknown variance σ^2 . [1]

To test for significance of β at the 5% level, we compare the absolute value of the observed test statistic $t(H_0: \beta = 0)$ i.e. 8.39 with the 5% critical value of t-distribution with 8 (=10 - 2) degrees

of freedom. As this is a two-sided test, we look for $t_{8;0.975} = 2.306$. Alternately, note that the p-value is given as 0.

Thus there is clear evidence that the value of β is significant at the 5% level. [1]

(ii) We consider regressing $\log_e y$ on x as the 'best' model. The reasons for this are as follows:

- Model 3 has the highest R^2 – though all are good
- Model 3 has the highest t values for the coefficients – though again all are good
- Model 3 has the lowest residual variance relative to the mean response
- Model 3 is the only one without a “large” residual
- There is some curvature in all the three plots. However, $\log_e y$ on x is slightly “straighter” than y on x

[Making at least 3 of the above points will earn full credit] [5]

(iii) Using (ii), we have $\log_e y = 5.413 - 0.322 x$

$$\Rightarrow y = \exp(5.413 - 0.322 x) \text{ or } 224.303 e^{-0.322 x}$$

[1]

(iv) From the expression in part (iii), inserting $x = 5$, $y = 224.303 e^{-0.322 * 5} = 44.8$.

This is in hundreds per square kilometer. So the estimate of population is 4,480 per square kilometer. [2]

(v) Prediction within the range of the data may be adequate, except perhaps near the upper end because of the tendency for curvature there. Extrapolation to values of x outside the data will, for similar reasons, be unreliable, and the regression model is likely to underestimate density. Where is the next city or town center? Interaction with that is very likely unless it is a long distance away. There may also be directional effects, i.e. densities changing more or less slowly according to the direction from the center.

[2]

[Total 16]

XXXXXXXXXXXXXXXXXXXX